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ORIGINAL RESEARCH PAPER

A Non-Destructive Pavement Evaluation for the Development of a Multi Distress Regression Model Integrated with the BACKCSA Model

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Abstract
This paper highlights the application of non-destructive tests (NDT) to evaluate the pavement for maintenance in a road network. The falling weight deflectometer (FWD) and the network survey vehicle (NSV) test data which is rational in terms of technical, financial, and environmental aspects have been used. The data using NDT devices were collected from a network of roads from Odisha state, India, and analyzed to assess the structural adequacy of the pavements. Further, using the FWD deflection data, the resilient moduli (M_R) of the pavements were back-calculated using a cuckoo search (BACKCSA) model developed as a part of this research work. The back-calculated layer moduli (M_R) was used as an input for developing the pavement condition index (PCI) prediction model. In addition, the other pavement distresses viz., international roughness index (IRI), rut depth, cracking, and raveling was also collected using the NSV. The data collected for three cycles with a minimum difference of one year between each cycle was used for the development of perdition models. The NSV data was used for developing the multilinear regression (MLR) prediction models for building a comprehensive pavement management system. This paper highlights the novel approach in arriving at the M_R, and incorporating the same in MLR models for predicting the maintenance strategy for a road network.

Keywords Deterioration models · Pavement evaluation · Back-calculation · PCI · FWD

1 Introduction

1.1 Background

There are two approaches for pavement evaluation; the first is by carrying out laboratory tests on the samples obtained from the field after cutting the cores. The above method is cumbersome, time-consuming, and detrimental to the pavement's layers. Cutting the cores quite frequently poses a threat to the safety of the crew, causes traffic delays, and is generally not accepted by road users. Furthermore, reproducing real-world outdoor conditions in laboratory testing is more complex. The second approach is to adopt the non-destructive testing (NDT) method to evaluate the pavement material properties [1–3]. The NDT refers to a variety of inspection procedures that allow investigators to analyze and collect data regarding the material, system, or component without permanently disturbing the pavement surface. Therefore, pavement condition data acquired from an NDT approach is considered to be a better approach than performing laboratory testing [1, 2, 4].

It is vital to examine the existing pavement condition, predict the progression of distresses and propose the maintenance treatment which is optimal to reduce the road user costs. The optimal maintenance strategy can be arrived by prioritizing the pavements on the basis of the existing condition [1–3]. It is necessary to carry out the functional and structural evaluation of the pavement to assess the existing pavement condition. The surface distresses data are collected through functional evaluation, while the pavement strength is assessed through structural evaluation. The functional distress data such as roughness, surface texture, cracking, potholes, raveling, and rutting can be collected quickly and accurately, using the network survey vehicle

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(NSV). However, among the functional distress data, the international roughness index (IRI) is considered as the crucial distress parameter for arriving at the maintenance strategy [5, 6].

In structural evaluation, the primary objective is to estimate the layer moduli using the falling weight deflectometer (FWD) data [1, 7–9]. From the FWD deflection data, the layer moduli are back-calculated using an iterative process until the differences between the measured and back-calculated deflection values are minimized [8, 10]. The back-calculation programs generally use the numerical integration subroutines for calculating pavement layer moduli using the FWD deflection data for known thicknesses and Poisson ratios [11–13]. As the back-calculation technique is a non-linear problem, a metaheuristic optimization approach is feasible as they converge to an optimum solution avoiding the local minima [10]. In addition, the back-calculated layer moduli values depend on the range of seed moduli given by the user as an input [11–13]. The available back-calculation layer moduli models have its own strengths and weaknesses, and no two models give the same set of results, precisely [14, 15]. Due to measurement errors and inadequacy of matching deflection basins, the back-calculated results are highly variable [15–17].

It is necessary to assess the surface condition of the pavement before assessing the structural evaluation of the pavement. Also, the functional distress parameters are used for developing the deterioration models which predicts the future condition of the pavement. The pavement deterioration models can be either deterministic or probabilistic. A probabilistic model incorporates the random variation, while a deterministic model does not. The deterministic models can be mechanistic-empirical or regression models. The mechanistic-empirical models, consider traffic, roughness, cracking, and rutting, while the regression models consider the relationship between the functional and structural distress parameters [18–21]. The surface condition of flexible pavement changes with time due to the combined effect of traffic and climate, affecting the pavement strength [22, 23]. Therefore, at least a three time-series data points are required to develop a pavement deterioration model using the regression methods [24, 25].

1.2 Review on Past Studies

The pavement condition index (PCI) method is the most widely used index to measure the present pavement condition based on the observed surface distresses using statistical analysis. It also indicates the structural integrity of the pavement. The deduct value curves given in ASTM D6433-18 considers 19 different distresses with different severity levels (low, medium and high) making the PCI application more challenging. In addition, it is also impractical to get the complete surface distresses data for a large network of roads, and unsafe to the inspection personnel. On the other hand, the image-based survey methods using NSV are faster and safer [18, 26].

Researchers have shown that IRI and PCI are the functions of pavement distresses with an acceptable statistical validity and the road agencies are following the same [27, 28]. Park et al. collected 63 data points from nine states in the US from 20 pavement sections using a power relationship between PCI and IRI [29]. Mactutis et al. collected 317 data points from the roads located near Reno, Nevada and established the relationship between IRI and PCI with a coefficient of determination, R^2 of 0.71 [30]. Arhin et al. developed the PCI model using IRI as a parameter for an urban area in Columbia and was found to be statistically significant with R^2 values between 0.56 and 0.82 [31]. Robbins et al. quantified the significance of IRI with respect to the increase in vehicle operating costs and considered the terminal IRI value as an indicator for arriving at the pavement maintenance measure [32]. The Mechanistic-Empirical Pavement Design (MEPDG) also, considers the terminal IRI as one of the main pavement structural design criteria [33]. The prediction models developed between PCI and IRI are not covering a wide range of data, and the accuracy of the models with limited data was fair [27]. In addition, the pavement develops multiple distresses, along with IRI due to traffic and climate. Therefore, the inclusion of multiple distresses for the development of a model leads to a rational PCI prediction model [34]. New technologies are now available to measure other important pavement distresses at the network level [35]. Hence, the structural capacity and surface friction can also be considered for the development of a successful PMS.

1.3 Objectives

In the present study, the deterministic pavement deterioration models using a multilinear regression method were developed [12, 36]. In addition, the back-calculated layer moduli are used in the development of the PCI model. A new back-calculation model was developed to determine the layered elastic moduli from FWD deflection data. To accommodate the data obtained from different FWDs', the back-calculation model, BACKCSA was developed using the cuckoo search (CSA) algorithm [4]. A 417 km of road network from Odisha state, India was considered for the network-level pavement evaluation using the data from NSV and FWD for developing the PCI model. The main objectives of the study were:

i. To arrive at the PCI value using deduct value method as per ASTM D6433-18

ii. To develop the performance prediction models for thin bituminous surfaces
iii. To determine the back-calculated layer moduli using BACKCSA
iv. To design the overlay for maintenance

2 Methodological Approach

The sixteen homogeneous road sections from the five districts of Odisha viz., Anugul, Bargarh, Cuttack, Deogarh, and Sambalpur were considered for this study. The derived data were divided into three groups using the k-means clustering algorithm [13]. Considering the traffic and pavement age as the variables, the IRI, deflection, cracking initiation, cracking progression, rutting initiation, rutting progression, and raveling initiation and raveling progression models were developed using a Minitab statistical tool. Further, the distress data were used for calculating the PCI values as per ASTM D6433-18 for predicting the pavement condition with respect to time [26, 37]. Finally, the BACKCSA model was developed using a meta-heuristic Cuckoo search algorithm (CSA). A detailed description of the methodology, along with the step-by-step procedure for the BACKCSA model, is presented in Fig. 1.

3 Data Collection

From the five districts of Odisha consisting of 16 pavement sections, a total of 417 km length having the traffic between 1500 and 2000 commercial vehicles per day (CVPD) were considered for this study. The data was collected during the years 2014, 2016, and 2018. The pavement composition consisted of semi-dense bituminous concrete (SDBC) as a surface layer, bituminous macadam (BM) as a binder layer, wet mix macadam (WMM) as a granular base, and a granular sub-base (GSB) as a sub-base layer, over a compacted sub-grade with a California bearing ratio (CBR) of 4%. The pavement sections were rehabilitated and upgraded during March 2013 and hence considered as the new pavements. Table 1 presents the road length and traffic in terms of CVPD for the road sections considered.

3.1 Pavement History and NSV Data

The pavement inventory data such as the road number, carriageway width, surface layer, shoulder type, and shoulder-widths were also collected. The road agency provided other information such as the pavement composition, layer thicknesses, and construction history (constructed during 2000 and rehabilitated during 2013). The pavement condition data

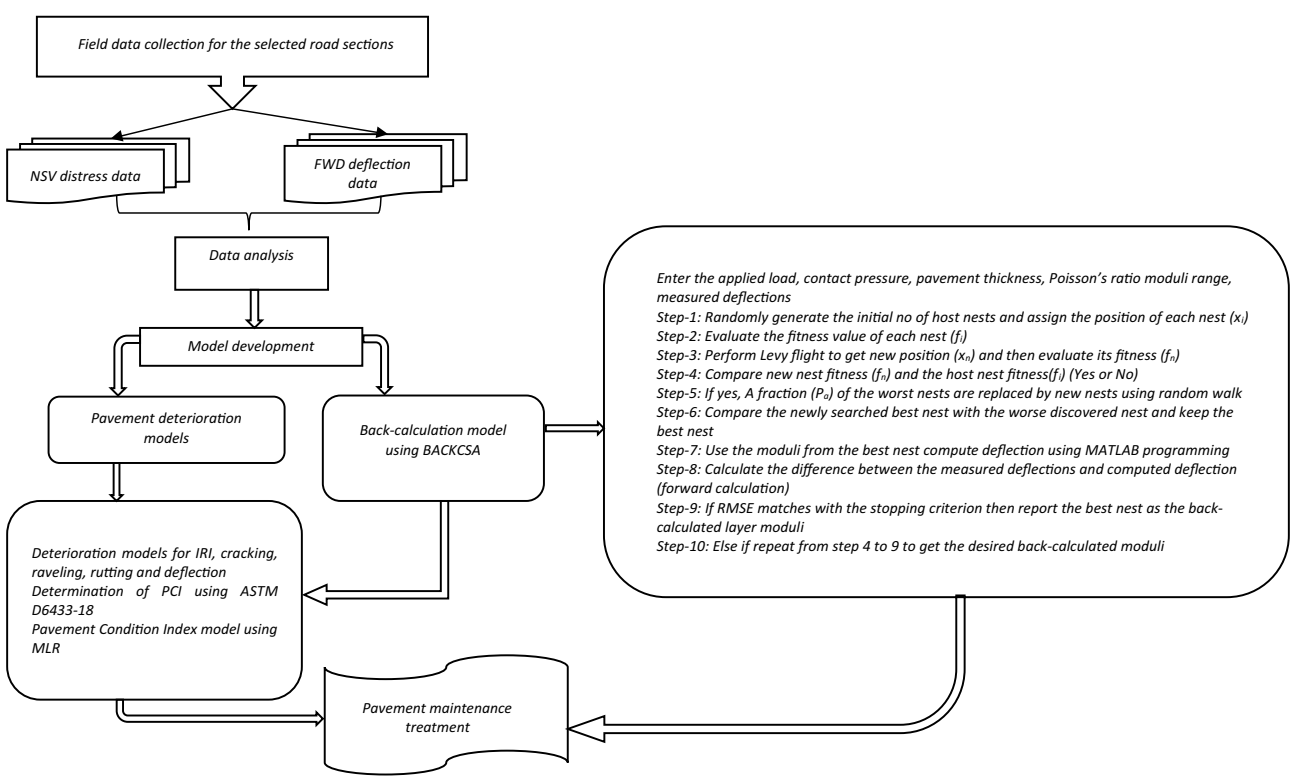


Fig. 1 Framework for the analysis of pavement evaluation data for a network-level PMS

viz., IRI, cracking, potholes, raveling, and rutting were collected using an automated NSV. Table 2 gives the sample distresses data obtained on 26th March 2018. Figure 2a and b show the NSV and FWD used for the data collection in Odisha state.

3.2 Pavement Deflection Data from FWD

The FWD, Dynatest model 8000 with a 300 mm diameter loading plate, was used for collecting the deflection data. The geophones were placed at 0, 200, 300, 600, 900, and 1200, 1500 mm from the center of the loading plate. Three

Table 1 Road length and CVPD on the road sections considered for the study

Division	Road ID	Start Chainage	End Chainage	Road length (km)	Traffic (CVPD)
Anugul	S-1	0	39.464	39.464	1536
Anugul	S-2	0	47.291	47.291	1612
Anugul	S-3	0	68.316	68.316	1675
Bargarh	S-4	0	17.549	17.549	1658
Bargarh	S-5	0	39.941	39.941	1576
Bargarh	S-6	0	7.863	7.863	1626
Cuttack	S-7	0	31.457	31.457	1854
Cuttack	S-8	0	11.107	11.107	1738
Cuttack	S-9	11.107	40.071	28.964	1844
Deogarh	S-10	0	7.013	7.013	1825
Deogarh	S-11	0	3.698	3.698	1954
Deogarh	S-12	0	42.565	42.565	1873
Sambalpur	S-13	45.271	68.125	22.854	1905
Sambalpur	S-14	0	17.907	17.907	1966
Sambalpur	S-15	70.2	83.655	13.455	1934
Sambalpur	S-16	0	17.907	17.907	1854

Table 2 NSV Sample data

Chainage (km)	International Roughness Index (m/km)	Cracking (%)	Raveling (%)	Rutting (mm)
0.000	4.5	0.0	0	4.5
1.000	3.2	5.9	0	3.8
2.000	3.6	0.0	0	3.7
3.000	4.6	8.2	0	4.1
4.000	6.5	1.5	0	3.8
5.000	5.4	0.0	0	3.1

(a) Network Survey Vehicle (b) FWD Dynatest 8000 used for the study

Fig. 2 **a** Network survey vehicle. **b** FWD Dynatest 8000 used for the study

Table 3 FWD specifications used for the study

Parameter	FWD
Impact load (kN)	40
Diameter of the loading plate (mm)	300
Number of geophones	9
Pulse duration (μs)	20–30
Drop height (mm)	100–600
Radial distances of geophones (mm)	0, 300, 450, 600, 900, 1200, 1500, 1800 and 2100

drops of mass were considered at each testing point with loading time ranging from 0.015 to 0.050 s, while the first drop was ignored, considering it to be a seating load. When the pavement was in good condition, the data were collected at 500 m intervals. However, when the pavement was in poor condition, the number of testing points was increased by reducing the testing interval. The deflection data obtained were normalized with respect to a standard load of 40 kN and corrected to a standard temperature of 35 °C. In addition, the seasonal correction was considered for sub-grade moisture. The FWD data analysis was carried out as per the guidelines set out by the Indian Roads Congress, IRC [38]. Table 3 shows the specifications of the FWD used in the study.

4 Data Analysis

The data collected from the 16 pavement sections from 5 different districts of Odisha were grouped into homogeneous clusters. A separate regression model was developed for each cluster to exhibit uniform characteristics and for getting better fitness [22]. However, the model for one cluster has been mentioned in the manuscript. The following section deals with the clustering method, prediction models, PCI model, and back-calculation model development with a detailed explanation.

4.1 K-means Clustering

Clustering is a process of grouping the data set with certain similarities. For unlabeled data, it is suggested to use k-means clustering as it is easy to implement, interpret, and secure [13]. The k-means is an established unsupervised machine learning algorithm for large datasets, and the steps adopted are given below:

Step-1: Partition of objects into k groups from the given data.
Step-2: Calculate the centroid (i.e., the mean for all variables) for each cluster.
Step-3: Determines the Euclidean distance of each object from the respective centroid (Eq. 1).
Step-4: Assigns each data element to its closest centroid.
Step-5: Computes the new cluster center for each cluster as the mean value of the elements in it.

Steps 3–5 have to be repeated until the cluster center calculation becomes constant and unchanged. The goodness of the partition is measured as the sum of squared distances.

For two n-dimensional data elements, $i = (x_{i1}, x_{i2}, \ldots, x_{in})$ and $j = (x_{j1}, x_{j2}, \ldots, x_{jn})$, a popular distance function is given in Eq. 1:

Euclidean Distance,

$$d_{ij} = \sqrt{\left[(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \cdots + (x_{in} - x_{jn})^2\right]}, \quad (1)$$

Road agencies need to segment a measurement series automatically, as the measurements from several homogeneous road sections will vary abruptly from one to another [39]. In the present study, the k-means clustering algorithm is used to delineate 417 km pavement sections using Minitab statistical tool. The number of data points considered in cluster-1, cluster-2, and cluster-3 were 163, 145, and 109, respectively. The variance for various iterations is given in Table 4, and the matrix plot for the distresses of all the pavement sections is shown in Fig. 3.

4.2 Development of Back-Calculation Model

4.2.1 Determination of Deflection Through Forward Calculation

Various soft wares developed based on Burmister's theory for calculating stresses, strains, and deflections, viz. BISAR, KENPAVE, CHEVRON, ELSA, etc. [17]. Generally, the bituminous concrete surface layer, bituminous concrete binder layer, unbound granular base layer, unbound granular sub-base layer, and a compacted subgrade layer are considered for the analysis. A 3-layered system with bituminous,

Table 4 K-means clustering groups

Variable	Cluster1	Cluster2	Cluster3	Grand centroid
RI	7.551	8.148	5.2	7.733
Cracking	0.937	0.251	0	0.718
Pothole	0.439	0.191	0	0.359
Raveling	0.525	0.288	0	0.449
Rutting	6.48	6.055	4.2	6.339
PCI	65.813	64.776	87.3	65.547
M_R	1450.051	1855.173	2330	1580.635

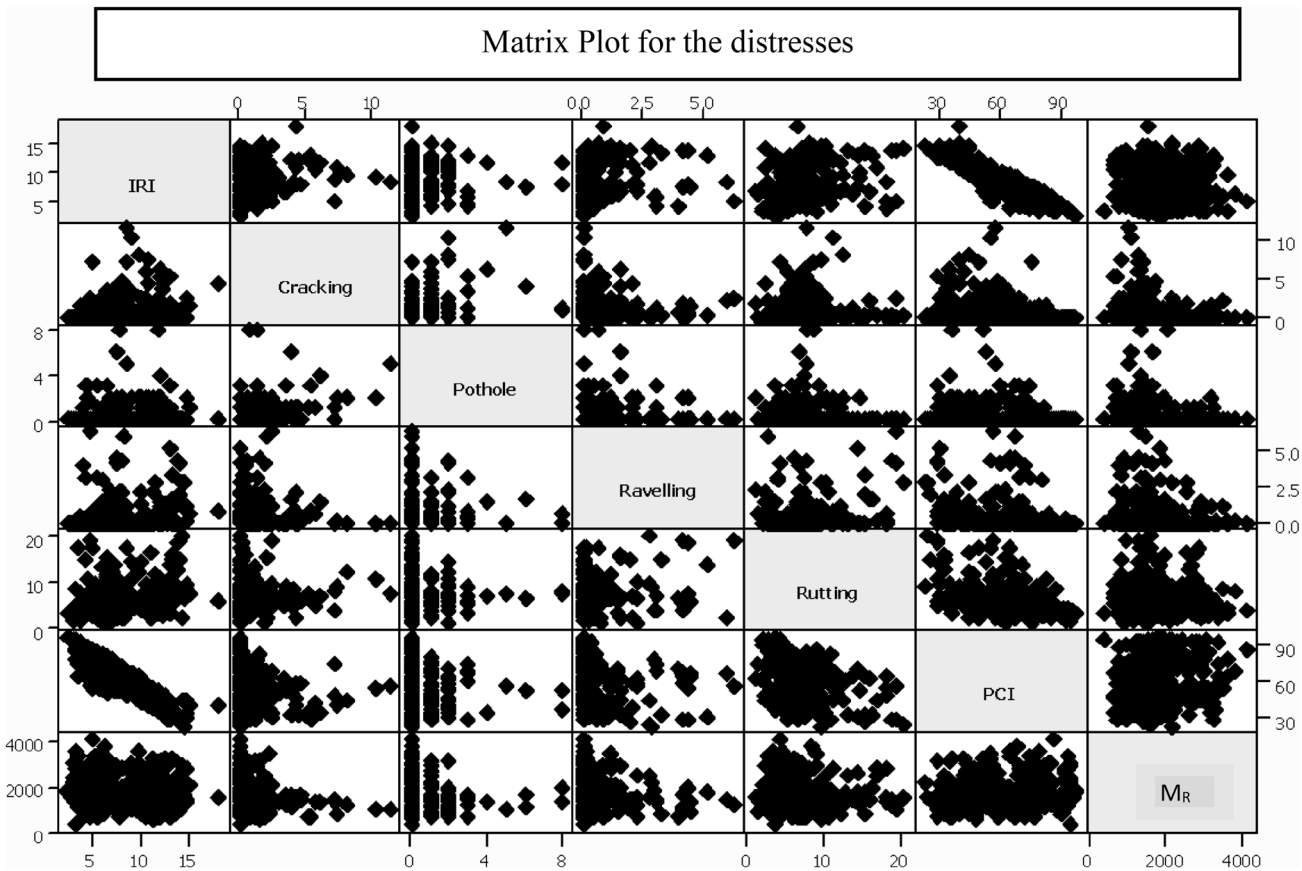


Fig. 3 Matrix plot for the pavement condition data

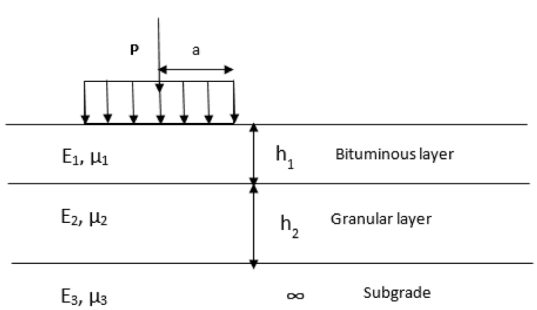


Fig. 4 Flexible pavement with three layers [18]

unbound granular, and compacted sub-grade layers is considered for all practical purposes, as shown in Fig. 4 [18].

The Burmister layered theory assumes that all the layers are numbered sequentially from top to bottom as '1' to 'n', with *n*th layer being an infinite elastic half-space. Following Burmister's theory, it is convenient to present the solutions of the linear elastic theory (LET), deflection (w_i) for layer i in terms of the following bi-harmonic function $\Phi_i\,(r, z)$:

$$w_i = \frac{1+\mu_i}{E_i}\left[2\left(1-\mu_i\right)\nabla^2\Phi_i - \frac{\partial^2\Phi_i}{\partial z^2}\right], \tag{2}$$

where μ_i is Poisson's ratio, E_i is Elastic modulus, z is the vertical coordinate measured from the top surface layer i.

By substituting the bi-harmonic function $\Phi_i\,(r, z)$ in Eq. (3), the deflection response can be rewritten in the form of Eq. (3):

$$w_i = \left[\int_0^{\infty}\left(\left[-A_I(m)\mathrm{m}e^{\mathrm{mz}} + B_I(m)\mathrm{m}e^{-\mathrm{mz}} + C_I(m)\left(2-4\mu_i-\mathrm{mz}\right) + D_I(m)\left(2-4\mu_i+\mathrm{mz}\right)e^{-\mathrm{mz}}\right]J_0(\mathrm{mr})\mathrm{m}d\mathrm{m}\right)\right], \tag{3}$$

where m is the integration parameter, $A_i(m)$, $B_i(m)$, $C_i(m)$, and $D_i(m)$ are the constants, J_0 is the Bessel function.

The simplified version of Burmister's bi-harmonic solution was proposed by Khazanovich and Wang which was used for the forward calculation in this study. In LET, the analysis is performed using the inverse Hankel transform, using numerical integration with the help of the Gaussian four-point Quadrature method [40]. The methodology followed for forward calculations is shown in Fig. 4 [40]. A computer program was developed using MATLAB to solve Eq. (4) with varying integration parameter 'm'. The system of equations to be solved are represented in the form of Eq. (4). For a three-layer system, there will be ten constants in $\{X\}$ and to solve these constants, ten equations are required. The entries in matrix M and the boundary conditions followed in this study are demonstrated by Khazanovich and Wang [40].

$$[M]\{X\} = \{Y\}, \tag{4}$$

where

$$M = \left[M_{11}00M_{21}M_{22}00M_{32}M_{33}\right], \tag{5}$$

$$\{X\} = \left\{A_1, B_1, C_1, D_1, A_2, B_2, C_2, D_2, B_3, D_3\right\}, \tag{6}$$

$$\{Y\} = \{\frac{\mathrm{qaJ_1(ma)}}{m^2}, 0, 0, 0, 0, 0, 0, 0, 0, 0\}. \tag{7}$$

The BACKCSA model considers the inputs of pavement layer thicknesses, Poisson's ratio, and normalized FWD deflection data. The deflection data from FWD has to be normalized to a standard load of 40 kN. The seed moduli ranges considered in the model are given in Table 5, and it was based on the IRC guidelines [24]. Different trials were carried out to determine the optimum set of parameters, and a number of solutions were obtained by varying one of the parameters while keeping the others constant [4]. Table 6 shows the descriptive statistics of the FWD data collected from the pavement sections.

Table 5 Seed moduli considered for the analysis

Layer	Lower bound (MPa)	Upper bound (MPa)	Poisson's ratio
Bituminous layer	750	5000	0.35
Granular layer	100	500	0.40
Subgrade	20	100	0.40

4.2.2 Back-Calculation Layer Moduli Using BACKCSA

The inputs for the BACKCSA model are the load applied, contact pressure, contact radius, pavement composition, Poisson's ratios, measured deflections (in excel format), number of sensors, radial distances, and the seed moduli range. Based on the number of nests, CSA assigns the position of each nest and evaluates the fitness value of each nest. It then compares the nests and keeps the optimal nest value (moduli) used for computing the theoretical deflections. This process is repeated until the difference between FWD and the calculated deflection values satisfies the convergence criteria [23, 28]. The simplified version of Burmister's bi-harmonic solution defined by Khazanovich and Wang [29] was used for the forward calculation of the deflections [29]. The well-known root means square error (RMSE) objective function considered in the back-calculation model was calculated using Eq. (2). RMSE value of less than 1% was chosen as the convergence criterion in the model Eq. (8) [12].

Table 6 Descriptive statistics of the measured deflections

Section	Statistical indicator	Measured deflections [μm]								
		0	300	450	600	900	1200	1500	1800	2100
		d_0	d_1	d_2	d_3	d_4	d_5	d_6	d_7	d_8
Cluster-1	Average	287	196	149	113	67	44	33	25	20
	Minimum	172	91	68	53	34	24	19	15	13
	Maximum	420	283	220	173	109	79	61	48	41
Cluster-2	Average	303	219	172	137	93	69	55	45	39
	Minimum	86	81	76	64	40	29	24	17	12
	Maximum	620	571	438	323	227	165	128	101	88
Cluster-3	Average	381	263	197	148	85	54	39	32	27
	Minimum	131	100	82	69	46	29	21	18	17
	Maximum	548	368	282	218	130	89	61	49	47

$$\mathrm{RMSE} = \left[\frac{\sum_{i=1}^{N}\left(W_{ci} - W_{\mathrm{FWD}i}\right)^2}{N}\right]^{0.5}, \tag{8}$$

where W_{ci} is the calculated deflection using back-calculated layer moduli (mm), $W_{\mathrm{FWD}i}$ is the measured deflection from FWD data (mm), N is the total number of sensors.

The FWD data collected was given as an input for each highway section to obtain the predicted layer moduli. The input (FWD deflection) and output (M_R) obtained from the BACKCSA model are taken from Vineesha et al. [6].

4.3 Remaining Service Life of the Pavements

After applying the temperature and seasonal corrections, the predicted layer moduli obtained using the BACKCSA model for each deflection point were saved in the database. The 15th percentile predicted layer moduli values were calculated and computed as given in Table 7 and were specified as the design moduli for designing an overlay. The critical tensile strain at the bottom of the bituminous layer and vertical compressive strain at the top of the subgrade were calculated using IITPAVE software based on LET theory. The remaining life was estimated from fatigue criteria using Eq. 15 given in the Indian Road Congress guidelines for the strengthening of flexible pavements [24].

$$N_f = 0.711 * 10^{-4} * 10^{4.84*\left(\frac{V_{\mathrm{be}}}{V_{\mathrm{a}}+V_{\mathrm{be}}}-0.69\right)} * \left[\frac{1}{\varepsilon_t}\right]^{3.89} * \left[\frac{1}{M_R}\right]^{0.854}, \tag{9}$$

where N_f is the fatigue life in million standard axles (msa), V_{be} is the percent volume of effective bitumen in the mix, V_{a} is the percent volume of air voids in the mix, ε_t is the tensile strain at the bottom of the bituminous layer, M_R is the back-calculated bituminous layer moduli from BACKCSA in MPa.

From Table 7, based on FWD data analysis, all the sections required the overlay in terms of bituminous concrete (BC). Figure 5 shows the proposed overlay thickness of 50 mm BC for one of the sections, S-2.

4.4 Development of Deterioration Prediction Models

For the development of distress prediction models, inventory data, functional data, and structural data were collected and compiled. The IRI, deflection, cracking initiation, cracking progression, rutting initiation, rutting progression, and raveling progression models are given through Eqs. 2–8, respectively, for cluster-1. The initial IRI for SDBC section was considered as 3.5 m/km for the analysis [22]. All the distress parameters considered for the analysis were statistically significant. For the other two clusters, similar distress prediction models were developed. The traffic in terms of million standard axles, expressed as cumulative standard axle (MSA) has been considered in the development of the models. The construction quality is considered as per HDM-4 for the road sections [41].

IRI progression model:

$$(\mathrm{IRI})_t = 0.156 * (\mathrm{IRI})_{t-1} - 0.616^{\mathrm{age}} + 0.5231 * \mathrm{MSA} + 1.185, n = 417, R^2 = 0.981, \mathrm{RMSE} = 0.605. \tag{10}$$

Deflection progression model:

$$(\mathrm{Def})_t = 0.018 * (\mathrm{Def})_{t-1} - \frac{0.0335}{\mathrm{Age}} + 0.0179 * \mathrm{MSA} + 0.146, n = 417, R^2 = 0.996, \mathrm{RMSE} = 0.005. \tag{11}$$

Cracking initiation model:

$$\mathrm{CRinitiation} = 2.56 * \exp * \left(-2.47\frac{\mathrm{MSA}}{3.2 * \left(\mathrm{D}_0\right)^{-0.63}}\right), n = 50, R^2 = 0.852, \mathrm{RMSE} = 0.147. \tag{12}$$

Cracking progression model:

$$(\mathrm{CR})_t = 0.024 * (\mathrm{CR})_{t-1} + 1.082^{\mathrm{age}} + 1.287 * \mathrm{MSA} + 0.542, n = 417, R^2 = 0.987, \mathrm{RMSE} = 0.495. \tag{13}$$

Raveling initiation model:

$$\mathrm{Ravinitiation} = 1.18 * \mathrm{MSA} * (\mathrm{CQ} + 1)^{-0.15}, n = 58, R^2 = 0.712, \mathrm{RMSE} = 0.256. \tag{14}$$

Rutting progression model:

$$(\mathrm{Rut})_t = 0.102 * (\mathrm{Rut})_{t-1} - \frac{4.746}{\mathrm{Age}} + 1.523 * \mathrm{MSA} + 0.446, n = 417, R^2 = 0.989, \mathrm{RMSE} = 0.597. \tag{15}$$

Raveling progression model:

$$(\mathrm{Rav})_t = 0.2 * (\mathrm{Rav})_{t-1} - \frac{0.0919}{\mathrm{Age}} + 0.0491 * \mathrm{MSA} + 0.556, n = 417, R^2 = 0.993, \mathrm{RMSE} = 0.015. \tag{16}$$

where IRI is the international roughness index at any time t (m/km), Crack is the cracking at any time t (%), M_R is the back-calculated bituminous layer moduli from BACKCSA, $\mathrm{CR}_{\mathrm{initiation}}$ is the cracking initiation (years), $\mathrm{Rav}_{\mathrm{initiation}}$ is the raveling initiation (years), Rav is the raveling at any time t (%), CQ is the construction quality, Age is the age of the pavement (years), MSA is the cumulative number of standard axles (millions).

Table 7 Back-calculated layer moduli using BACKCSA and remaining service life

Road ID	Total length (km)	Thickness (mm)		Design traffic (msa)	Fatigue life before overlay (msa)	BACKCSA layer moduli			Overlay thickness (mm)	Fatigue life after overlay (msa)
		Bituminous	Granular			Bituminous E1 (MPa)	Granular E2 (MPa)	Subgrade E3 (MPa)		
S-1	39.464	75	350	30.48	10.43	1502	321	63	70	35.91
S-2	47.291	75	350	31.99	15.82	1779	368	67	50	32.34
S-3	68.316	75	350	37.21	8.73	1399	306	76	80	41.58
S-4	17.549	75	350	34.99	17.95	1243	359	60	60	36.99
S-5	39.941	75	350	31.28	23.07	1063	379	84	50	34.70
S-6	7.863	75	350	34.25	17.98	1872	388	86	50	38.25
S-7	31.457	75	450	36.79	20.39	1274	384	86	50	37.93
S-8	11.107	75	450	34.49	9.80	1751	318	67	65	36.05
S-9	28.964	75	450	36.59	7.23	1243	284	63	80	37.20
S-10	7.013	75	450	33.24	16.69	1724	374	76	50	36.90
S-11	3.698	75	450	38.78	18.78	1403	380	83	55	42.07
S-12	42.565	75	500	37.17	12.49	1721	346	85	60	39.90
S-13	22.854	75	500	37.81	15.72	1962	372	81	50	39.59
S-14	17.907	75	500	39.02	10.79	1063	315	73	70	39.27
S-15	13.455	75	500	38.38	22.81	1208	391	76	50	40.94
S-16	17.907	75	500	34.81	9.64	1883	317	67	65	37.98

4.5 Pavement Condition Index (PCI) Models

The PCI is expressed by a numerical value based on the rating system on a scale of 0–100, with '0' corresponding to the worst possible condition, while '100' corresponding to the best condition of a pavement section [26, 36].

4.5.1 Calculation of Pavement Condition Index (PCI) as per ASTM D6433-18

Considering the distress density, the PCI was calculated using the deduct value method as per ASTM D6433-18 [26]. The curves given in ASTM D6433-18 were digitized using the Engauge digitizer program to obtain the mathematical equations and to calculate the deduct value for all pavement distresses considered. The relationship between the deduct value and the distress density (%) was plotted for all the distresses, and Fig. 6 shows an example plot for raveling. The plot for raveling distress density is shown in Fig. 6a, while the digitized plot is shown in Fig. 6b. The mathematical model to calculate the deduct value for medium severity raveling density is given in Eq. 17. Similarly, the mathematical models for all other distress at low, medium and high severity were obtained by digitizing the distress curves given in ASTM D6433-18 and were used for calculating the PCI.

$$y = 36.002x^4 - 125.44x^3 + 161.9x^2 - 83.256x + 19.552,\ R^2 = 0.998, \tag{17}$$

where y is the deduct value and x is the logarithmic value of raveling density in percentage.

4.5.2 Development of PCI Model Using Multilinear Regression

By substituting the field data in the ASTM D6433-18, the PCI values were obtained and using this value a multilinear regression PCI model was developed as given in Eq. 18.

$$\begin{aligned} \text{PCI} &= 100 - (4.264 * \text{IRI}) - (1.203 * \text{Crack}) - (0.734 * \text{Rav}) \\ &\quad - (0.883 * \text{Rut}) + \left(0.0265 * M_R\right), \\ &\quad n = 417, R^2 = 0.91,\ \text{RMSE} = 6.34. \end{aligned} \tag{18}$$

5 Model Validation

This section deals with the validation of the developed BACKCSA model. The developed PCI model was validated through a statistical analysis using an independent t-test.

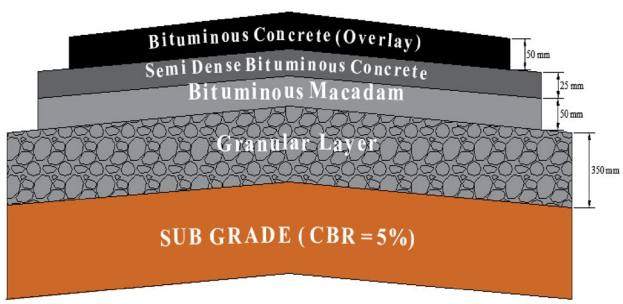
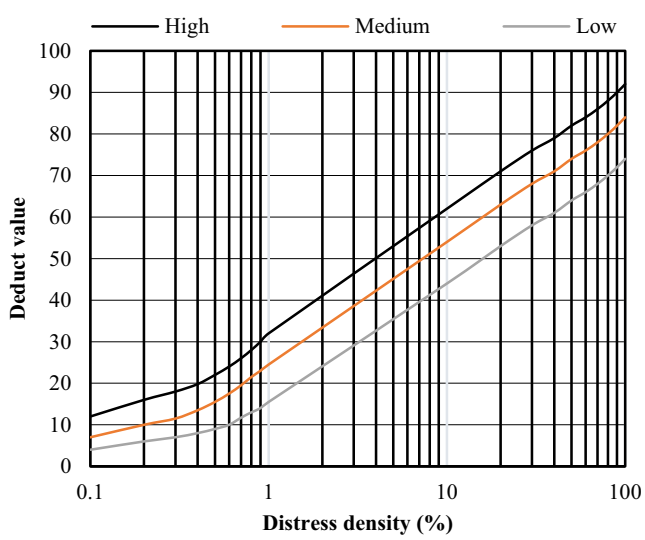
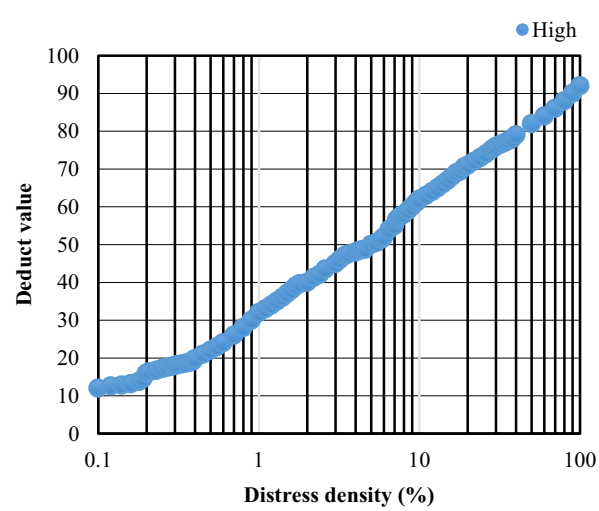


Fig. 5 Proposed overlay for section, S-2



(a) Raveling chart from ASTM D6433-18



(b) Raveling chart from digitizer

Fig. 6 a Raveling chart from ASTM D6433-18. b Raveling chart from digitizer

5.1 PCI Progression Model

The PCI progression with respect to time (years) for cluster 1 is shown in Fig. 7. Similar trends were observed for cluster 2 and 3 (not shown in this paper). Two interventions, intervention-1 and intervention-2, were considered and plotted to demonstrate the PCI trend with respect to time (years). The intervention-1 or intervention-2 can be selected based on the availability of budget with respect to the length of the road for maintenance within the network. In intervention-1, for example, the terminal PCI is considered as 50 (PCI@50) and the pavement deterioration shall be allowed to such an extent, where the PCI will not be less than or equal to 50. Similarly, in intervention-2, the terminal PCI is considered as 60 (PCI@60) and the pavement deterioration shall be allowed to such an extent, where the PCI will not be less than or equal to 60. In intervention-2, the budget required will be higher than that of intervention-1, as the pavement will be maintained at a higher level of service, compared to intervention-1.

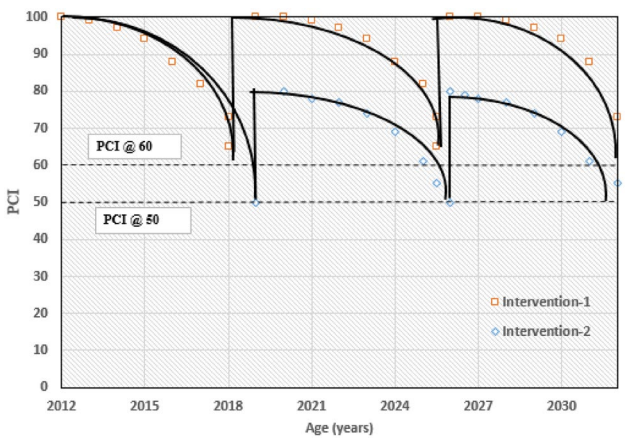


Fig. 7 PCI progression for cluster-1

5.2 Statistical Analysis for Goodness of Fit of the PCI Model

The statistical analysis was done using Minitab to obtain the goodness of fit of the PCI model and is graphically represented through Fig. 8a–d. From Fig. 8a, it is observed that the data set is approximately normally distributed with a minimum variance and follows an ordinary least square (OLS) calculation, as shown in Fig. 8b. As a result, the PCI model generates an unbiased coefficient estimate with minimum variance. From Fig. 8c, the data is skewed at the center and follows a normal distribution. The residuals are evenly distributed and are consistent with respect to the centerline for all the observations, as shown in Fig. 8d.

5.3 Relationship Between the Observed and the Predicted PCI Data

A linear model regression equation was considered to find the correlation between the observed and the predicted PCI values, and the coefficient of determination, R^2 was 0.856. This indicated a good correlation between the observed and the predicted values. The observed and predicted data are similar, with a coefficient of determination of 0.856. The relationship between predicted PCI values obtained from Eq. 10 and the observed PCI values obtained from the deduct value method for cluster-1 is shown in Fig. 9. Similar trends were observed for the other clusters too.

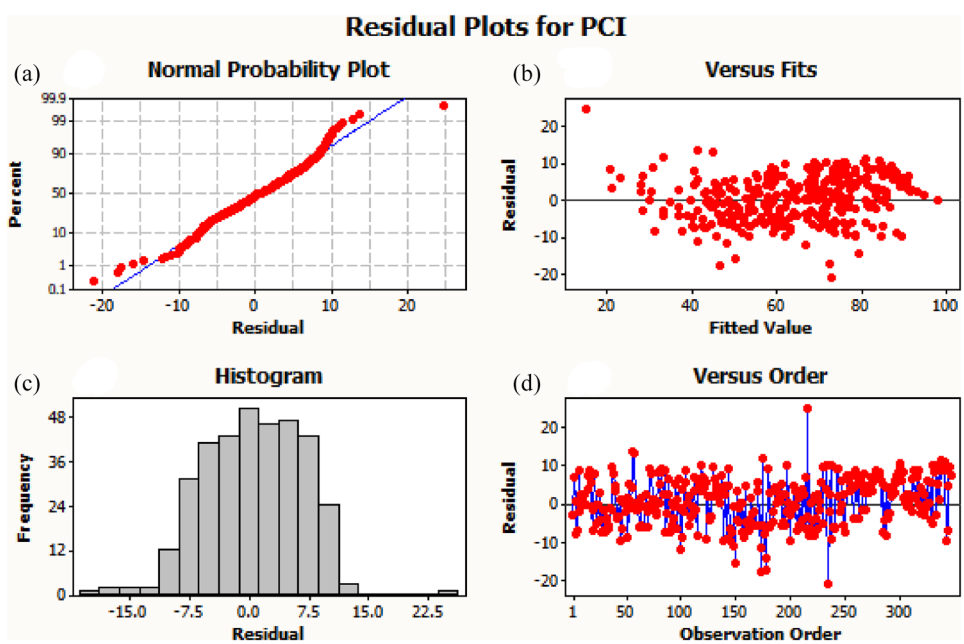


Fig. 8 Residual plots for PCI model from Minitab

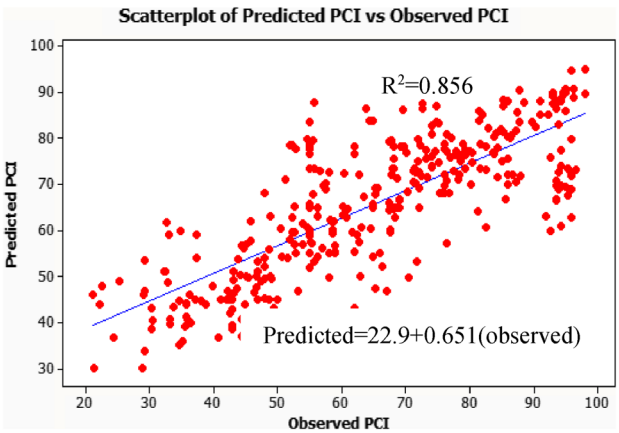


Fig. 9 Relationship between predicted and observed PCI values for cluster-1

Table 8 Independent *t*-test for observed and predicted PCI values

	PCI observed	PCI predicted
Mean	65.51	64.94
Variance	215.26	370.28
Observations	417	417
Hypothesized Mean Difference	0	
d_f	649	
t Stat	2.8	
P($T \leq t$) two-tail	0.005	

5.4 Statistical Analysis Using Independent *t*-Test

A statistical analysis was carried out using Minitab to perform the independent t-test between observed and the predicted PCI values to show that the variances are not equal. The null hypothesis and an alternative hypothesis considered is given below:

Null hypothesis: No significant difference between predicted and the observed PCI values.
Alternate hypothesis: There is a significant difference between predicted and the observed PCI values.

The output obtained from Minitab for an independent t-test is given in Table 8. From the table, it can be seen that the mean values are 65.51 and 64.94 for the observed and predicted PCI values, respectively. In addition, from the table it can be seen that the variances are 215.26 and 370.28 for the observed and the predicted PCI values, respectively. If the p value is less than 0.05 (significance level), and the difference between the variances are statistically significant, then the null hypothesis fails to get rejected. Hence, from the output given in Table 8, it can be inferred that the null hypothesis fails to get rejected and the predicted PCI values have a good correlation with the observed PCI values.

6 Conclusions

The NDT approach is preferable over an expensive, time-consuming, and unsafe conventional destructive testing approach. The NDT data helps analyze the structural condition of the pavement and helps optimize the maintenance activities. The developed multi distress linear regression approach enables the decision-makers to assess the existing pavement condition and to arrive at the maintenance measures using the terminal PCI as an intervention criterion. The developed deterioration models are applicable for the roads with traffic ranging from 1500 to 2000 commercial vehicles per day, with a tropical climate and medium rainfall conditions. The models can be used for district roads, rural roads and a few of the state highways fulfilling the above-mentioned conditions. One of the most significant benefits of using the BACKCSA model over the other back-calculation models is its ability to handle deflection data from any FWD equipment, in general.

1. Since collecting structural and functional data at the network level of road projects is expensive, the progression models developed in this study from three cycles' data can be used to select the optimum maintenance measure for the pavement's life cycle while avoiding the costs of repeated data collection.
2. It is highly cost-effective to homogenize the pavement sections within the network based on IRI, traffic, pavement composition, and environment to improve prediction accuracy in the developed deterioration models.
3. A multi distress regression model predicting PCI was proposed. The model is a function of IRI, cracking, rutting, raveling, back-calculated layer moduli, MSA, construction quality, age of the pavement. This model yielded an R^2 value of 0.91.
4. In conjunction with the BACKCSA model, the FWD deflections were used to evaluate the strength of the pavement structures. From the data analysis, the structural deficiencies of the pavements arrived at various locations of SHs. The study illustrated that the timely evaluation of pavement sections is essential in selecting the appropriate pavement maintenance measures.
5. Generally, the substantial benefit of using the BACKCSA model over the other back-calculation models is its ability to handle deflection data from any FWD equipment. The layer moduli obtained from the BACKCSA model can calculate the remaining service life of the pavements and design the overlays. Furthermore, this model can be

linked with the PAMS to serve as an integral part of estimating an effective network-level maintenance strategy.

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Declarations

Conflict of Interest No potential conflicts of interest were declared by the authors.

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